

HOW TO WRITE A RESEARCH PAPER

Step 1: Preparation Before Writing

- Emphasizes the importance of having **all materials ready before starting to write**, comparing it to cooking where ingredients must be prepped in advance.
- Critical items include:
 - **Findings** fully analyzed and finalized
 - **Figures and tables** completed and organized
- Advises against starting to write the paper before having these core elements ready, as writing should be the **final 10% of the project**.
- Warns that starting to write too early leads to confusion and inefficiency.

Step 2: Write the Paper “Inside Out”

- Stuckler asserts that trying to write the paper linearly from Introduction → Methods → Results → Conclusion is a **major mistake** and causes writer’s block.
- Instead, start with the **Methods section**, which is often the easiest to write and helps build momentum.
- After Methods, write the **Results section**, focusing on describing your tables, figures, and key findings in a clear and logical order.
- The goal is to create a **coherent narrative** centered around your data to show readers and journal editors the story your research tells.
- This reverse approach helps avoid getting stuck and accelerates progress.

Step 3: Write the Conclusion Early

- Contrary to common practice, write the **Conclusion before the Introduction**.
- The Conclusion typically has a **formulaic structure** with multiple clearly defined components:
 1. **Recap your key findings** succinctly without detailed explanation.

2. List and discuss the **limitations** of your study transparently. This is a strategic move to **preempt reviewer criticisms** and demonstrate awareness of your study's weaknesses. Stuckler recommends listing limitations clearly and simply (e.g., "first," "second," "third").
 3. Explain the **implications for future research** based on your findings.
 4. Outline any potential **policy or practical interventions** relevant to your field. (May not apply to all fields.)
- This structured conclusion provides a strong foundation for the rest of the paper.

Step 4: Write the Introduction Last

- The Introduction is often the most difficult part and a major stumbling block.
- It has a **three-part structure**:
 1. **Establish relevance**: Explain why the topic matters now, highlighting its current importance broadly or within the field.
 2. **Identify gaps**: Discuss the weaknesses or missing elements in existing literature, supported by a brief literature review citing prior studies.
 3. **Present your study's value**: Clearly articulate how your research addresses these gaps and the unique contribution it makes.
- Writing the introduction last allows you to tailor it precisely after having clearly defined your methods, results, and conclusions.

Step	Task Description	Purpose/Benefit
1. Preparation	Finalize findings, figures, tables	Avoid confusion, have core content ready
2. Methods	Write Methods section first	Easiest section, builds confidence
3. Results	Describe tables, figures, and key findings	Create coherent narrative, avoid writer's block

Step	Task Description	Purpose/Benefit
4. Conclusion	Recap findings, list limitations, discuss implications	Preempt criticism, clarify impact and future work
5. Introduction	Explain relevance, identify gaps, state study value	Frame the study, highlight unique contribution

Summary Table: Paper Writing Sequence

Key Insights and Takeaways

- **Writing order matters:** Writing "inside out" (Methods → Results → Conclusion → Introduction) is far more efficient than linear writing.
- **Preparation is crucial:** Completing all data analysis, figures, and tables before writing saves time and reduces frustration.
- **Transparent limitations:** Openly discussing study limitations strengthens the paper by anticipating reviewer concerns.
- **Formulaic conclusion:** Following a structured conclusion format streamlines writing and enhances clarity.
- **Iterative refinement:** Accept imperfect first drafts and polish over time to overcome perfectionism and writer’s block.
- **Introduction last:** Writing the introduction last ensures it is targeted and relevant after the rest of the paper is drafted.

Literature Review Introduction and Importance

What is a Literature Review?

- A literature review is likened to catching up with an old friend after many years, where you learn about key developments, challenges, and opportunities from their life.
- Similarly, a literature review synthesizes **key concepts, current knowledge, gaps, debates, and potential contributions** within a specific academic field or topic.
- It provides a **snapshot of the state of the art** and highlights areas where further research is needed.

Step 1: Dive Straight In – Starting Your Search

- Avoid “paralysis by analysis” by starting immediately with exploratory searches, even if you know little about the topic.
- Use **Google Scholar** for initial searches to identify top papers and key contributors in your field.
- Example given: searching "inequality and health" to retrieve highly cited or recent influential papers.
- Download PDFs of about **10-15 relevant papers** to begin the reading and summarizing process.
- *Note:* Systematic reviews require different techniques; a separate video is referenced for that method.

Step 2: The Strip Method – Extracting Key Information

- Create a Word document with citations listed for each paper to avoid plagiarism and maintain organization.
- Carefully read each paper, especially abstracts and key sections, and **extract (strip out) relevant points and findings**.
- This initial extraction will look messy but is essential for gathering raw material.
- Include **direct quotes with quotation marks** to preserve original meaning and attribution.
- Summarize:
 - What the authors did

- Their findings
- Suggestions for future research
- Contradictory evidence if present
- Example: A systematic review on inequality and health found evidence of a **causal link** between inequality and health outcomes but noted some contradictory findings due to measurement differences.
- This phase also helps you begin to identify **patterns and structure** emerging from the literature.

Step 3: Develop a Conceptual Framework – Structuring Your Review

- The conceptual framework forms the **backbone or skeleton** of your literature review, organizing the information into a coherent structure.
- Various approaches to structuring include:
 - **Chronological order:** useful for historical or developmental topics.
 - **Problem-to-solution:** progressing from describing an issue to discussing proposed solutions.
 - **Levels of analysis:** for example, individual-level evidence followed by population-level evidence.
 - **Thematic breakdown:** grouping by key themes or debates within the field.
- Example from the inequality and health topic:
 - Papers showing **causal links** versus papers showing **associations without causation**.
- After deciding on the structure, use the **snowball method** to expand your search by following references cited in key papers to find more relevant studies and deepen your review.

Step 4: Writing with the PEER System – Paragraph Structure

- The **PEER system** defines the anatomy of a strong academic paragraph:
 - **P (Point):** Make one clear, concise claim or topic sentence.
 - **E (Evidence/Example):** Provide supporting studies or data.
 - **E (Explain):** Interpret the evidence, explaining its significance.

- **R (Repeat/Link):** Reinforce the point or transition smoothly to the next idea.
- Example paragraph on inequality and health:
 - **Point:** Several studies found a causal relationship between inequality and ill health.
 - **Evidence:** Citing Wilkinson and Pickett, Jamison et al.
 - **Explanation:** These studies meet criteria for causality (temporality, strength, specificity).
 - **Repeat:** Conclude that there is strong evidence for causality at broad geographic levels.
- Paragraphs should be clear, focused, and logically linked, building the argument progressively.

Step 5: Conclusion and Introduction Writing Order

- Contrary to common practice, write the **conclusion before the introduction**.
- The conclusion should include four key components:
 1. **Summary of findings:** Recap the state of knowledge and key insights from the review.
 2. **Limitations:** Discuss weaknesses or gaps either in your review or in the literature itself (e.g., limited papers, methodological weaknesses). This transparency helps anticipate criticism and improve credibility.
 3. **Future research directions:** Suggest areas needing further investigation based on identified gaps.
 4. **Policy/practice implications:** If applicable, propose how findings might inform action or decision-making.
- Writing the conclusion first is easier once the evidence is synthesized and provides a clear foundation to frame the introduction.

Crafting the Introduction

- The introduction should engage the reader by explaining **why the topic matters now**—the urgency or significance of the debate.
- Typical structure:

1. **Context and relevance:** Example—rising inequality’s impact on avoidable deaths and suffering.
 2. **Gaps and debates:** Highlight contested areas or unresolved questions in the field, foreshadowing the conceptual framework.
 3. **Purpose and scope:** Define what the literature review specifically addresses, identifying the motivation behind it.
- Writing the introduction last ensures clarity and coherence since the review’s findings and structure are fully formed.

Final Advice and Support Offer

- Following these five steps—dive in, strip key info, develop a framework, write with PEER, and conclude before introducing—will help students produce **high-quality, efficient literature reviews**.
- Professor Stuckler offers additional **one-to-one coaching sessions** for students struggling with academic writing, aiming to help them overcome obstacles and accelerate their graduate careers.

Summary Table of the Five Steps for Literature Reviews

Step Number	Name	Description	Key Actions
1	Dive Straight In	Begin with Google Scholar searches to find key papers even if unfamiliar with topic	Search, download 10-15 papers
2	Strip Method	Extract key points, findings, quotes from papers, keeping citations for plagiarism avoidance	Copy citations, highlight findings, summarize
3	Conceptual Framework	Organize findings into a coherent structure (chronological, thematic, problem-solution, etc.)	Map papers, develop themes, use snowball method
4	PEER Writing System	Write paragraphs with clear points, evidence, explanation, and linking/repetition	Compose focused paragraphs

Step Number	Name	Description	Key Actions
5	Conclusion then Intro	Write conclusion summarizing findings, limitations, future research and implications; then intro	Draft conclusion; write introduction last

Introduction to Manuscripts and Importance of Publishing

- A **manuscript** traditionally refers to any document written by hand or typewritten, as opposed to mechanically printed or reproduced automatically.
- Publishing a research paper is crucial because:
 - It **enhances public understanding** of a research question.
 - It **establishes the author as an expert** in their field of knowledge.
 - Peer-reviewed publications provide **evidence for evaluating the merit of research funding requests**.

Initial Steps to Publish a Manuscript in a Journal

- To publish a manuscript:
 1. **Select the journal** where you want to submit your manuscript.
 2. Since it is your first submission to this journal, **register yourself** on the journal's platform.
 3. During registration, provide all required information such as:
 - Name
 - Affiliation
 - Other necessary details
 4. Complete registration by clicking the finish button.

Submission Process Overview

- After registration, go to the journal's main page and:

1. Click on **“Submit the paper”**.
 2. Login using your **username and password**.
 3. Access your **author dashboard** and start submitting your manuscript.
- The submission process is divided into multiple steps:
 - **Step 1:** Enter the **title** and **abstract** of the manuscript.
 - **Step 2:** Upload the **manuscript files**, including tables and figures.
 - **Step 3:** Add **co-authors** and their affiliations.

Detailed Submission Steps

- **Step 4:** Provide additional manuscript details such as:
 - Number of words
 - Number of figures
 - Number of tables
- **Step 5:** Review all entered information carefully.
 - The system generates a PDF version of the manuscript for proofreading.
 - Add **keywords** relevant to your research.
 - Attach a **cover letter** for submission.
- Authors must proofread the generated PDF and correct any mistakes before final submission.
- Once satisfied, the author clicks **submit** to officially send the manuscript to the journal.

Post-Submission Process and Timeline

- After submission:
 - The author receives a **confirmation email with a manuscript ID**, confirming the journal has received the paper.
 - The **editor reviews the manuscript** for suitability.
- If the manuscript is:
 - **Suitable:** It is sent for **peer review**, which can take several months depending on the journal.

Step No.	Action	Details/Requirements
1	Registration	Provide name, affiliation, and other data
2	Login and start submission	Use username and password
3	Enter manuscript title and abstract	First page of submission
4	Upload manuscript files	Includes tables and figures
5	Add co-authors and their affiliations	
6	Enter manuscript details	Number of words, figures, tables
7	Proofread PDF generated by the system	Add keywords and cover letter
8	Final submission	Submit after proofreading
9	Editor review	Manuscript checked for suitability
10	Peer review (if suitable)	Duration varies by journal
11	Acceptance and final proofreading	Author receives acceptance email
12	Publication	Final version published within a few days

- **Not suitable:** It is **rejected**.
- Upon acceptance:
 - The author receives an **acceptance email**.
 - The author proofreads and finalizes the manuscript version for publication.
 - The paper is then **published within a few days** after final proofreading.

Summary Table: Manuscript Submission Process

Key Insights

- **Manuscript submission is a stepwise, systematic process** requiring careful input and proofreading at every stage.
- **Registration and profile completion are mandatory** before manuscript submission.
- **Peer review is a critical phase** that determines the acceptance or rejection of the manuscript.
- Timelines for review and publication vary by journal but can take **several months** for review and a few days post-acceptance for publication.

[00:00]

Introduction and Motivation

- The speaker introduces the video aimed at helping beginner researchers understand the process of publishing research work.
- Motivation stems from the speaker's own initial confusion and lack of guidance regarding where and how to publish research findings.
- The video intends to raise awareness and provide clarity on the paper writing and publishing process, applicable across all research disciplines.

[00:40]

Overview of Paper Writing and Publishing Stages

- Research publication involves several broad stages, which the speaker will explain in detail.
- The first crucial stage is manuscript writing or preparation, where researchers transcribe their concepts, theories, or findings, supported by collected data.
- This stage demands brainstorming, dedication, motivation, attention to detail, and multiple rounds of editing until the manuscript is finalized.

[01:28]

Manuscript Composition and Components

- Once finalized, the draft becomes a manuscript, generally consisting of:
 - Preliminary material
 - Body of the text

- Reference list
- For beginners, focus is placed on the preliminary material, which includes:
 - Title: Should be concise, comprehensive, attractive, and specifically highlight the paper's focus.
 - Authors: Author order indicates contribution; the first author contributed the most, others might be supervisors, co-researchers, or collaborators.
 - Affiliations: Institutions where research was conducted; multiple affiliations are possible.
 - Corresponding author: The primary contact for journal communication, listed on the front page or as a footnote.
 - Keywords: Four to six selected words to facilitate retrieval of the research, analogous to search keywords in online platforms.

[04:28]

Body of Manuscript and Key Sections

- The body includes:
 - Abstract
 - Introduction
 - Materials and Methods
 - Results and Discussion
 - Conclusion and Recommendations
 - Tables and Figures
- The reference list is the final major component.
- A good manuscript typically has the following characteristics:
 - Original and plagiarism-free content
 - Correct research methodology, including sampling design and statistical tools
 - Contribution to existing scientific knowledge
 - Ethical considerations addressed
 - Proper, effective writing style

[05:58]

Journal Selection: Factors to Consider

- **Selecting an appropriate journal requires in-depth research and careful consideration.**
- **Important factors to look for in a journal:**
 - **Relevant to the researcher's discipline or area**
 - **Peer-reviewed by experts**
 - **Has an International Standard Serial Number (ISSN) for electronic and print formats**
 - **Indexed in a globally or nationally recognized database**
 - **Included in the UGC CARE list (especially relevant for Indian researchers)**
 - **Journal has a history of at least 8 to 10 volumes (indicates stability and credibility)**
 - **Authentic publishing house**
 - **Transparent fee structure (some charge, some do not)**

[07:31]

Journal Selection: Factors to Avoid

- **Researchers should avoid journals that:**
 - **Have poor quality or lack rigorous peer review**
 - **Do not follow standard publishing practices**
 - **Are not included in recognized indexing or UGC CARE lists**
 - **Are categorized as predatory or "D3" journals**
 - **Demand immediate payments upon submission**
 - **Promise unrealistically fast publication (within a few days)**

[09:04]

Manuscript Preparation According to Journal Guidelines

- **After selecting a journal, the manuscript must be formatted per the journal's author guidelines found on the journal's website.**
- **Key formatting aspects to adjust:**

- Font type, size, and spacing
- In-text citation style (e.g., author-year system or numbered references in superscript or brackets)
- Reference list formatting and completeness (all cited in text must appear in the list and vice versa)
- Preparation and numbering of tables and figures, and their placement (within the text or at the end)
- Each journal has unique formatting requirements, so careful adherence is essential.

[11:30]

Manuscript Submission Process

- Researchers must follow instructions on:
 - File format (single Word file, PDF, or section-wise files)
 - Submission mode: either via email or through a journal's online portal (requiring registration and login)
- Along with the manuscript, additional documents may be required, including:
 - Cover letter: Explains purpose of submission, addressed to the editor
 - Declaration letter: Confirms manuscript has not been published elsewhere
 - Suggested reviewers: Some journals ask authors to recommend 2-3 expert reviewers
 - Funding information: Discloses financial support for the research
 - Author contributions: Brief professional details or roles of contributing authors
 - Copyright form: Signed by all authors to transfer copyright, mandatory either at submission or post-acceptance
 - Conflict of interest and ethical statements, if applicable

[14:21]

Post-submission: Evaluation and Decision

- After submission, the journal acknowledges receipt with an acknowledgement letter.
- The manuscript undergoes peer review by experts, which may take months or longer.

- Possible editorial decisions communicated to the corresponding author include:
 - Acceptance for publication (with an acceptance letter)
 - Rejection (with a non-acceptance letter)
 - Revision requests: Reviewers may ask for clarifications or additional information; authors revise accordingly.
- Revisions improve manuscript quality, and this process may repeat multiple times before final acceptance.

[16:38]

Final Stage: Publication and Galley Proof

- Once accepted, the manuscript is professionally edited and formatted according to the journal's style.
- Authors receive a galley proof—a near-final version for proofreading.
- Authors must carefully check for:
 - Spelling errors
 - Missing pages or content
 - Correct placement of figures, tables, and text
- Any errors or omissions must be reported immediately for correction.
- After approval of the galley proof by both parties, the manuscript is officially published as a research paper.

[17:20]

Conclusion and Final Remarks

- The video emphasizes understanding the entire publication process from manuscript inception to final publication.
- The speaker expresses hope that the video aids beginner researchers in navigating these stages effectively.
- The speaker encourages viewers to like, share, comment, and subscribe if they find the video helpful.

Summary Table: Key Stages of Research Paper Publication

Stage	Description	Key Points
Manuscript Preparation	Writing and editing the research manuscript	Brainstorming, multiple revisions, components: title, authors, abstract, body, references
Journal Selection	Choosing an appropriate journal	Peer review, ISSN, indexing, UGC CARE, journal reputation, fees
Manuscript Formatting	Adjusting manuscript to journal-specific guidelines	Citation style, reference list, font, tables/figures placement
Submission	Sending manuscript and required documents via email or online portal	File formats, cover letter, declaration, reviewer suggestions, copyright forms
Peer Review & Decision	Evaluation by experts; acceptance, rejection, or revision requests	Can take months, revisions improve quality
Publication & Galley Proof	Final editing, proofreading, and formal publication	Authors review galley proofs, report errors, manuscript officially published

Key Insights

- Manuscript writing is the foundation of research publication, requiring focus and iterative refinement.
- Journal selection critically influences the visibility and credibility of the published work.
- Strict adherence to journal-specific formatting and submission guidelines enhances acceptance chances.
- Peer review is a vital quality control mechanism, often involving multiple rounds of revision.
- The publication process is lengthy and requires patience but culminates in dissemination of validated scientific knowledge.

Question	Answer
What is a manuscript?	A handwritten or typewritten document submitted for publication.
Why publish a research paper?	To enhance public understanding, establish expertise, and support funding requests.
What is the first step in manuscript submission?	Register yourself on the journal's platform.
How do you submit a manuscript?	Login, fill required details, upload files, add co-authors, proofread, and submit.
What happens after submission?	Editor reviews, followed by peer review if suitable.
How long does peer review take?	It varies; can take several months depending on the journal.
What happens after manuscript acceptance?	Final proofreading and publication within a few days.

Publishing process next

paper writing and publishing process

Overview of Paper Writing and Publishing Stages

- Research publication involves several broad stages, which the speaker will explain in detail.
- The first crucial stage is **manuscript writing or preparation**, where researchers transcribe their concepts, theories, or findings, supported by collected data.
- This stage demands **brainstorming, dedication, motivation, attention to detail**, and multiple rounds of editing until the manuscript is finalized.

Manuscript Composition and Components

- Once finalized, the draft becomes a **manuscript**, generally consisting of:
 - **Preliminary material**
 - **Body of the text**
 - **Reference list**
- For beginners, focus is placed on the **preliminary material**, which includes:
 - **Title:** Should be concise, comprehensive, attractive, and specifically highlight the paper's focus.
 - **Authors:** Author order indicates contribution; the first author contributed the most, others might be supervisors, co-researchers, or collaborators.
 - **Affiliations:** Institutions where research was conducted; multiple affiliations are possible.
 - **Corresponding author:** The primary contact for journal communication, listed on the front page or as a footnote.
 - **Keywords:** Four to six selected words to facilitate retrieval of the research, analogous to search keywords in online platforms.

Body of Manuscript and Key Sections

- The **body** includes:

- Abstract
- Introduction
- Materials and Methods
- Results and Discussion
- Conclusion and Recommendations
- Tables and Figures
- The **reference list** is the final major component.
- A **good manuscript** typically has the following characteristics:
 - Original and plagiarism-free content
 - Correct research methodology, including sampling design and statistical tools
 - Contribution to existing scientific knowledge
 - Ethical considerations addressed
 - Proper, effective writing style

Journal Selection: Factors to Consider

- Selecting an appropriate journal requires **in-depth research** and careful consideration.
- Important factors to look for in a journal:
 - Relevant to the researcher's discipline or area
 - **Peer-reviewed** by experts
 - Has an **International Standard Serial Number (ISSN)** for electronic and print formats
 - Indexed in a globally or nationally recognized database
 - Included in the **UGC CARE list** (especially relevant for Indian researchers)
 - Journal has a history of at least **8 to 10 volumes** (indicates stability and credibility)
 - Authentic publishing house
 - Transparent fee structure (some charge, some do not)

Journal Selection: Factors to Avoid

- Researchers should avoid journals that:
 - Have **poor quality** or lack rigorous peer review
 - Do not follow **standard publishing practices**
 - Are not included in recognized indexing or UGC CARE lists
 - Are categorized as **predatory or “D3” journals**
 - Demand **immediate payments upon submission**
 - Promise **unrealistically fast publication** (within a few days)

Manuscript Preparation According to Journal Guidelines

- After selecting a journal, the manuscript must be formatted per the journal’s **author guidelines** found on the journal’s website.
- Key formatting aspects to adjust:
 - Font type, size, and spacing
 - In-text citation style (e.g., author-year system or numbered references in superscript or brackets)
 - Reference list formatting and completeness (all cited in text must appear in the list and vice versa)
 - Preparation and numbering of tables and figures, and their placement (within the text or at the end)
- Each journal has unique formatting requirements, so **careful adherence** is essential.

Manuscript Submission Process

- Researchers must follow instructions on:
 - File format (single Word file, PDF, or section-wise files)
 - Submission mode: either via email or through a journal’s online portal (requiring registration and login)

- Along with the manuscript, additional documents may be required, including:
 - **Cover letter:** Explains purpose of submission, addressed to the editor
 - **Declaration letter:** Confirms manuscript has not been published elsewhere
 - **Suggested reviewers:** Some journals ask authors to recommend 2-3 expert reviewers
 - **Funding information:** Discloses financial support for the research
 - **Author contributions:** Brief professional details or roles of contributing authors
 - **Copyright form:** Signed by all authors to transfer copyright, mandatory either at submission or post-acceptance
 - **Conflict of interest and ethical statements,** if applicable

Post-submission: Evaluation and Decision

- After submission, the journal acknowledges receipt with an **acknowledgement letter**.
- The manuscript undergoes peer review by experts, which may take months or longer.
- Possible editorial decisions communicated to the corresponding author include:
 - **Acceptance for publication** (with an acceptance letter)
 - **Rejection** (with a non-acceptance letter)
 - **Revision requests:** Reviewers may ask for clarifications or additional information; authors revise accordingly.
- **Revisions improve manuscript quality**, and this process may repeat multiple times before final acceptance.

Final Stage: Publication and Galley Proof

- Once accepted, the manuscript is professionally edited and formatted according to the journal's style.
- Authors receive a **galley proof**—a near-final version for proofreading.
- Authors must carefully check for:
 - Spelling errors

- Missing pages or content
 - Correct placement of figures, tables, and text
- Any errors or omissions must be reported immediately for correction.
- After approval of the galley proof by both parties, the manuscript is **officially published as a research paper**.

Summary Table: Key Stages of Research Paper Publication

Stage	Description	Key Points
Manuscript Preparation	Writing and editing the research manuscript	Brainstorming, multiple revisions, components: title, authors, abstract, body, references
Journal Selection	Choosing an appropriate journal	Peer review, ISSN, indexing, UGC CARE, journal reputation, fees
Manuscript Formatting	Adjusting manuscript to journal-specific guidelines	Citation style, reference list, font, tables/figures placement
Submission	Sending manuscript and required documents via email or online portal	File formats, cover letter, declaration, reviewer suggestions, copyright forms
Peer Review & Decision	Evaluation by experts; acceptance, rejection, or revision requests	Can take months, revisions improve quality
Publication & Galley Proof	Final editing, proofreading, and formal publication	Authors review galley proofs, report errors, manuscript officially published

Key Insights

- **Manuscript writing is the foundation of research publication, requiring focus and iterative refinement.**
- **Journal selection critically influences the visibility and credibility of the published work.**

- **Strict adherence to journal-specific formatting and submission guidelines enhances acceptance chances.**
- **Peer review is a vital quality control mechanism, often involving multiple rounds of revision.**
- **The publication process is lengthy and requires patience but culminates in dissemination of validated scientific knowledge.**

Manuscript Writing targets researchers who have completed their scientific project or thesis analysis and now wish to communicate their findings to the broader scientific community. It emphasizes that after investing significant effort in data collection and analysis, many lose momentum at the writing stage. However, **writing and publishing your results is a crucial responsibility**, ensuring your research influences practice, guidelines, or community impact. The session aims to help viewers:

- Organize ideas logically and sequentially within a predefined manuscript structure.
- Furnish additional information to finalize the manuscript.
- Articulate ideas effectively to communicate the intended message clearly.

A **manuscript** is defined broadly as any document submitted for publication—including music or poetry—but here, the focus is on **scientific manuscripts**. These can be:

- Original research articles
- Review articles
- Correspondences
- Perspectives
- Editorials

The lecture advises starting manuscript writing only after you have:

- Final results presented in tables and figures.
- 2 to 3 central ideas or key recommendations to emphasize.
- An outline of the argument you intend to present.

This initial preparation sets the stage for structured writing.

IMRD structure of scientific manuscripts:

Section	Purpose
Introduction	Explains the rationale and why the research question was pursued.
Methods	Describes what was done to answer the research question in detail.
Results	Presents the findings without interpretation.
Discussion	Interprets the findings, putting them into context and meaning.

An example article titled *Impact and Operational Feasibility of TrueNat MTB Rif under India's RNTCP* is referenced to illustrate these sections.

In the **Introduction**, the goal is to **focus the reader's attention** amid many distractions and provide:

- Background of the research area.
- Identification of knowledge gaps.
- The rationale for the study.
- Specific objectives the research seeks to answer.

This section sets the context and prepares the reader for the study's importance.

The **Methods section** explains the research process in **clear, stepwise detail**, providing:

- Study setting and context (important for international readers).
- Study design.
- Operational definitions.
- Sampling strategy and sample size.
- Study procedures (data collection, investigations).
- Data management (collection and analysis).
- Ethical considerations and human participant protections.

Methods vary by study design but must provide **transparency and reproducibility** to ensure credibility.

The **Results section** is typically the briefest, focused solely on **what was found** without interpretation. Results may be presented using:

- Text
- Tables
- Figures
- Graphs
- Images

Authors must avoid repeating information and adhere strictly to journal guidelines on the number and types of displays. **Interpretation is reserved for the Discussion section.**

The **Discussion section** is described as the most important and intellectually demanding part, where the researcher:

- Summarizes key findings briefly (not more than three lines).
- Offers interpretations and inferences.
- Compares findings with existing literature (while avoiding overemphasis on comparisons).

Key advice: Balance between comparing with literature and emphasizing your own interpretations—this section reveals the researcher’s independent thinking.

Within the Discussion, also address:

- **Strengths** of the study, such as novel methodologies, design, or unique contexts.
- **Limitations**, reported honestly and with commentary on how they may affect results.

Following this, provide:

- **Conclusions:** The central take-home message aligned with primary objectives.
- **Recommendations:** Practical advice for policymakers, clinicians, or other stakeholders based on conclusions.

Staying focused on the 2 to 3 central ideas through results, discussion, conclusions, and recommendations helps readers assimilate findings effectively.

Discussion section structure:

- Summary of key findings.
- Interpretation and inference.
- Comparison with existing literature.
- Strengths and limitations.
- Conclusions.
- Recommendations.

Abstract

is described as a miniature manuscript and the most commonly read part. It can be:

- **Structured abstract:** Divided into Introduction, Methods, Results, and Conclusions.
- **Unstructured abstract:** A summary without explicit subheadings.
- **Plain language summary:** For lay readers or the general public.

Abstracts typically range from **100 to 350 words** and require careful crafting since many readers base their decision to read further on the abstract.

Keywords are critical for indexing and ensuring discoverability in search engines. Selecting correct and relevant keywords:

- Helps the article be found by targeted searches.
- Increases the likelihood of being read and cited.

References must credit all sources used, focusing on relevant and recent literature. Important points include:

- Use of reference management software to ease formatting.
- Awareness of different referencing styles such as Vancouver or Harvard.
- Proper in-text citations linked with reference lists at the manuscript end.

IMRD structure and **Argument Matrix**, which:

- Organizes 2 or 3 central ideas logically across each manuscript section.
- Ensures continuity and flow of ideas from Introduction through Methods, Results, and Discussion.
- Helps maintain a **linear and coherent narrative**.

Example of argument matrix flow:

Manuscript Section	Idea 1	Idea 2	Idea 3
Introduction	Present rationale for Idea 1	Present rationale for Idea 2	Present rationale for Idea 3
Methods	Methods addressing Idea 1	Methods addressing Idea 2	Methods addressing Idea 3
Results	Findings on Idea 1	Findings on Idea 2	Findings on Idea 3
Discussion	Interpretation of Idea 1	Interpretation of Idea 2	Interpretation of Idea 3

This approach enhances clarity and reader comprehension.

Optional manuscript sections may include:

- **Highlights:** 5 to 8 bullet points summarizing key messages.
- **Research in Context:** Specifies what is known and what the study adds to the field.

theses and dissertations:

- A thesis is an extended, university-mandated document for postgraduate or doctoral degrees.
- It is **primary research led by the scholar**, guided by a mentor.
- Bound by university-specific guidelines.
- Can be converted into a manuscript for journal publication.
- Manuscripts and theses differ in length, word limits, target audience, and sometimes structure (thesis may include additional sections).

finalizing the manuscript

by adding essential supplementary information:

- Title page (authors, affiliations, word count).
- Cover letter.
- Funding source declaration.
- Conflict of interest statement.
- Author contributions (adhering to ICMJE criteria).
- Appendices and supplementary files.
- Acknowledgments for contributors not qualifying for authorship.

Additional thesis-specific sections may include:

- Table of contents.
- List of abbreviations.
- List of tables and figures.

The lecture stresses **plagiarism checks** before submission to ensure originality and ethical compliance.

To **articulate content effectively**, the lecturer recommends:

- Use **simple, specific, and necessary words**.
- Maintain **consistent terminology**.
- Use **inclusive and non-offensive language**, as some journals require explicit statements about this.
- Use **precise word order** and **avoid passive voice**.
- Keep sentences **concise**, avoiding long or convoluted structures.
- Submit manuscripts for **scientific language editing** (grammar, spelling, punctuation, structure) by professionals or specialized software.

Regarding **formatting**, it is essential to strictly follow journal guidelines on:

- Font size and style.
- Indentation and spacing.
- Heading and paragraph styles.

The lecture highlights the "**six S's**" of **scientific writing** as guiding principles:

Six S's	Description
Simple	Use straightforward language.
Short	Keep sentences and text concise.
Structured	Follow a clear organizational format.
Sequential	Logical flow of ideas.
Strong	Use assertive, confident statements.
Specific	Be precise and detailed.

The final steps recommended include:

- Writing and rewriting drafts.
- Multiple rounds of self-review.
- Seeking peer or expert feedback.
- Revising based on constructive criticism until submission-ready.

Key take-home messages:

- Identify and focus on **2 to 3 central ideas** around which the manuscript is woven.
- Use the **IMRD structure** to organize content.
- Fit ideas logically and sequentially in each section.
- Apply **scientific writing style tips** for clarity and engagement.

- Adhere to **university or journal guidelines** for thesis or manuscript respectively.
- Recognize that **scientific writing improves with practice**, and patience is key.
- Embrace the cycle of **write, review, revise** to achieve a polished final manuscript.

WRITING PEER REVIEW

How Peer Reviewers Are Found

This is a common question — editors find reviewers by:

- Looking at **who published similar papers** recently
 - Checking **who is cited** in the submitted paper
 - Using databases like **Publons** or **Web of Science**
 - Personal **professional networks**
 - Researchers can also **volunteer** to review on platforms like **Publons (Clarivate)**
-

Communities by Field

- **PubMed / MEDLINE** — medical and life sciences
- **IEEE Xplore** — engineering and technology
- **SSRN** — social sciences, economics, law
- **PhilPapers** — philosophy
- **ERIC** — education research

Introduction to Writing a Peer Review Report

Step 1: Overview

- Start the review by thanking the editors for the opportunity to review the paper, which is a customary academic practice.
- Provide a **brief summary of the paper's topic and aims** to demonstrate understanding and context.

- Example given: For a paper titled *“Virtual Reality and Digital Education: A Network of Forance Perspective on Effective Use,”* an appropriate summary might be:
 - The paper aims to **enhance understanding of immersive technologies in digital learning**.
 - It reports results from an **interview study** and develops a **theoretical framework** to help educators integrate virtual reality into curricula.

Step 2: State Your Expertise

- Reviews are always **anonymous**, but it is important that editors and authors understand the reviewer's **background and expertise**.
- Include 1-2 sentences describing your relevant research experience to establish credibility.
- Mention how you will structure the review and if any aspects will be de-emphasized due to your expertise.
- Example:
 - “Having contributed to several studies in virtual reality myself, I consider my expertise significant. However, as I primarily use quantitative methods, I will focus less on methodology and ask editors to seek other reviewers’ opinions on this.”
- This transparency helps editors contextualize your comments and complements the overall review process.

Step 3: Structuring the Main Sections of the Review

- Two common ways to structure the main body of a review:
 1. **Chronological approach**: Provide feedback on each section of the paper (introduction, literature review, methods, results, etc.).
 2. **Thematic approach** (recommended): Organize the review around **major points of criticism** instead of following the paper’s structure.
- The video strongly advises against the chronological method, stating it is not the approach used by experienced reviewers.

- Suggested structure involves identifying about **3 to 6 main points of critique**, each given roughly equal attention.
- Example of main points:
 - Theoretical motivation of the study
 - Lack of transparency in method description
 - Missing theoretical contribution

Step 4: Constructive Criticism

- For each main point, structure your critique into three parts:
 1. **Positives** — Start by noting what you liked about the paper related to this point. This balances the review and acknowledges the authors' efforts.
 2. **Negatives** — Clearly state your criticisms, specifying exact parts of the text to help authors locate issues.
 3. **Suggestions for Improvement** — The most crucial element; offer concrete, actionable advice to improve the paper.
- Suggestions can include:
 - Recommending additional literature
 - Proposing stronger arguments
 - Suggesting methodological changes
- Emphasizes the importance of remaining **objective** and **constructive** rather than harsh or overly negative.

Step 5: Minor Points

- Minor points are smaller issues noticed during reading, such as:
 - Linguistic inconsistencies
 - Citation errors

- Create a brief list of these, including page references. Example:

Minor Issue	Page Number	Comment
“Effect” should be “effect on”	2	Typographical error
“There should be” → “there are”	3	Grammar correction
Missing page number in direct quote	4	Citation detail incomplete

- If too many minor errors exist, do not list all individually; instead, make a **general comment** recommending professional proofreading.
- Clarifies that the reviewer is not a proofreader but should point out persistent issues.

Step 6: Including References in Your Review

- A **professional review includes a short reference list** citing sources used to support critique or recommendations.
- This practice helps authors easily access the literature you mention and facilitates addressing your points.
- Including references elevates the review from mediocre to very good by demonstrating scholarly rigor and thoroughness.
- Encourages reviewers to make it as easy as possible for authors to implement suggestions.

Step 7: Recommendation

- The reviewer must provide a recommendation about the paper’s fate:
 - Reject
 - Major revisions required
 - Minor revisions required
 - Accept as is
- The recommendation is typically **not included directly in the review text** to simplify the editor’s task of summarizing multiple reviews.

- The final decision lies with the editor, not the reviewer.
- Even if recommending rejection, the reviewer’s critique should be constructive and helpful since authors still benefit from the feedback.
- Example concluding paragraph that avoids stating the recommendation explicitly:

“In summary, this paper addresses an interesting case for implementing virtual reality in digital education. Unfortunately, in its current state, it shows fundamental weaknesses such as in theoretical motivation, transparency, methodological implementation, and lack of theoretical contribution. I hope my critique helps the authors improve their work and find something useful in this review. Best of luck in developing the study further.”

Summary Table: Seven Steps to Writing a Peer Review

Step	Description	Key Elements
1	Overview	Thank editors, brief paper summary
2	State Your Expertise	Background, focus of review
3	Structure Main Sections	Use 3-6 major points of critique, not chronological
4	Constructive Criticism	Positives, negatives, and suggestions for each point
5	Minor Points	List minor issues, recommend proofreading if many
6	References	Cite sources supporting your critique
7	Recommendation	Provide decision privately, avoid explicit in text